

SOFTWARE ENGINEERING

Lab-03: Component Modelling & Architectural Pattern Selection

Problem Statement 53: Freelance Content Creator Escrow Platform

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Scenario

The Freelance Content Creator Escrow Platform lets a Content Creator submit draft media deliverables for a project milestone, which the system automatically watermarks before a Client Sponsor reviews them. The sponsor can approve the milestone — releasing the escrowed payment to the creator via an external Payment Gateway — or request revisions with comments, or raise a dispute that freezes payment and escalates to platform support. Both parties are notified by email/in-app alert on every status change. The core challenge is coordinating several independent workflows (submission, review, payment, disputes, notifications) around a shared escrow balance while keeping payment data secure and the review experience responsive.

Architecture Selection

We chose Microservices Architecture for the Freelance Content Creator Escrow Platform.

The platform is organized as a set of independently deployable services identified from the use-case model: a Draft Submission & Watermarking Service, a Milestone Workflow Manager, an Escrow Payment Service, a Notification Service, a Dispute Management Service, and a shared Milestone & Asset Data Store. Each service exposes a well-defined provided interface and consumes required interfaces from the services it depends on (see accompanying Component Diagram).

Reason 1 – Independent Scaling & Fault Isolation

Draft watermarking is CPU/media-intensive and spikes when many creators submit deliverables, while escrow payment processing is transaction-critical but comparatively low-volume. Microservices let each concern scale independently and fail without cascading — if the watermarking service is overloaded or crashes, milestone approval, payment release, and notifications continue to function.

Reason 2 – Alignment with Independent Business Workflows

Submission/watermarking, review/approval, payment release, dispute handling, and notifications are distinct business capabilities with different release cadences (e.g., payment-gateway compliance changes should not force a redeploy of the notification pipeline). Microservices map cleanly onto these bounded contexts, matching the FR-001–FR-005 requirements identified in Lab 1/2, each of which is owned by a single service.

Security Advantage

Isolating the Escrow Payment Service as its own microservice lets sensitive payment credentials and transaction data (encrypted per NFR-002, TLS 1.2+ in transit / AES-256 at rest) be confined to a single, narrowly-scoped service with its own access controls and audit logging — rather than being reachable from the broader application codebase — reducing the blast radius of a compromise in any other component.

Performance Benefit

Because the Draft Submission & Watermarking Service is decoupled from the Milestone Workflow Manager, watermarking (bounded by NFR-001 at under 5 seconds) can be scaled horizontally with additional instances during peak upload periods without adding load to, or being blocked by, milestone review, payment, or notification traffic — keeping the sponsor-facing review experience responsive.

UML Component Diagram

Escrow Platform - UML Component Diagram

Freelance Content Creator Escrow Platform | Microservices Architecture | SRN: PES1UG24AM363

